

Assignment 23

Q1. Create a program to display database records one by one using ResultSet.

Java Code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class Q1_DisplayRecords {

    private static final String URL = "jdbc:mysql://localhost:3306/SIT_CSE";
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public static void main(String[] args) {
        String selectQuery = "SELECT * FROM records";

        try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery(selectQuery)) {

            System.out.println("===== DISPLAYING RECORDS ONE BY
ONE =====");
            int count = 1;

            // Navigating row by row using next()
            while (rs.next()) {
                System.out.println("--- Record #" + count + " ---");
                System.out.println("ID      : " + rs.getInt("id"));
                System.out.println("Title   : " + rs.getString("title"));
                System.out.println("Category : " + rs.getString("category"));
                System.out.println();
                count++;
            }
        }
    }
}
```

```

    }

    } catch (SQLException e) {
        System.out.println("Database Error: " + e.getMessage());
    }
}
}

```

SQL Code:

```
USE SIT_CSE;
```

```
CREATE TABLE IF NOT EXISTS records (
    id INT PRIMARY KEY,
    title VARCHAR(100),
    category VARCHAR(50)
);
```

```
INSERT INTO records (id, title, category) VALUES
    (1, 'Database Systems', 'Textbook'),
    (2, 'Java Programming', 'Guide'),
    (3, 'Data Structures', 'Reference')
ON DUPLICATE KEY UPDATE title = VALUES(title);
```

Output:

```

===== DISPLAYING RECORDS ONE BY ONE =====
--- Record #1 ---
ID       : 1
Title    : Database Systems
Category : Textbook

--- Record #2 ---
ID       : 2
Title    : Java Programming
Category : Guide

--- Record #3 ---
ID       : 3
Title    : Data Structures
Category : Reference

```

Q2. Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department and salary sequentially

Java Code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class Q2_EmployeeResultSetNav {

    private static final String URL = "jdbc:mysql://localhost:3306/SIT_CSE";
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public static void main(String[] args) {
        String selectQuery = "SELECT Eid, Ename, Department, Salary FROM
employee";

        // Creating a scrollable ResultSet statement for flexible navigation
demonstration
        try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
            Statement st = con.createStatement(
                ResultSet.TYPE_SCROLL_INSENSITIVE,
                ResultSet.CONCUR_READ_ONLY
            );
            ResultSet rs = st.executeQuery(selectQuery)) {

            System.out.println("===== EMPLOYEE RECORDS
(SEQUENTIAL DISPLAY) =====");

            // Traversing sequentially using next()
```

```

while (rs.next()) {
    int id = rs.getInt("Eid");
    String name = rs.getString("Ename");
    String dept = rs.getString("Department");
    double salary = rs.getDouble("Salary");

    System.out.println("Employee ID : " + id);
    System.out.println("Name      : " + name);
    System.out.println("Department : " + dept);
    System.out.println("Salary    : " + salary);
    System.out.println("-----");
}

} catch (SQLException e) {
    System.out.println("Database Error: " + e.getMessage());
}
}
}

```

SQL Code:

```
USE SIT_CSE;
```

```

CREATE TABLE IF NOT EXISTS employee (
    Eid INT PRIMARY KEY,
    Ename VARCHAR(100),
    Department VARCHAR(50),
    Salary DOUBLE
);

```

```

INSERT INTO employee (Eid, Ename, Department, Salary) VALUES
    (101, 'Rahul', 'IT', 50000.0),
    (102, 'Aman', 'HR', 45000.0),
    (103, 'Rohit', 'Finance', 55000.0),
    (104, 'Akshat', 'IT', 65000.0)
ON DUPLICATE KEY UPDATE Ename = VALUES(Ename);

```

Output:

===== EMPLOYEE RECORDS (SEQUENTIAL DISPLAY) =====

Employee ID : 101
Name : Rahul
Department : IT
Salary : 50000.0

Employee ID : 102
Name : Aman
Department : HR
Salary : 45000.0

Employee ID : 103
Name : Rohit
Department : Finance
Salary : 55000.0

Employee ID : 104
Name : Akshat
Department : IT
Salary : 65000.0
